

SIMATS ENGINEERING

ASSESSMENT TOOL 3: Comparative Analysis

QUESTION 1 – DETECTION PERFORMANCE COMPARISON

Given Data

Model	True Positives (TP)	False Positives (FP)	False Negatives (FN)
A – Viola-Jones	80	20	30
B – YOLO	90	15	20

(a) Calculate Precision and Recall

Formula for Precision

Formula for Recall

Model A – Viola-Jones

Precision:

Recall:

Model B – YOLO

Precision:

Recall:

Comparison

Model	Precision	Recall
Viola-Jones	80.00%	72.73%
YOLO	85.71%	81.82%

(b) Evaluation

YOLO achieves higher precision and recall than Viola-Jones.

- Precision improvement = $85.71\% - 80\% = 5.71$ percentage points
- Recall improvement = $81.82\% - 72.73\% = 9.09$ percentage points

Final Answer

Model B – YOLO is better for object detection.

Justification

YOLO provides both **higher precision (85.71%)** and **higher recall (81.82%)** compared with Viola-Jones, which achieves 80% precision and 72.73% recall. Therefore, based on the calculated performance measures, **YOLO demonstrates better overall object detection performance.**

QUESTION 3 – OPTICAL FLOW MAGNITUDE COMPARISON

Given Data

Region Method A Method B

R1	5	8
R2	6	10
R3	5	9

(a) Calculate Average Motion Magnitude

Formula

Method A

Method B

Comparison

Method Average Motion Magnitude

Method A 5.33 pixels/frame

Method B 9.00 pixels/frame

(b) Evaluation

Method B has a higher average optical-flow magnitude:

Therefore, Method B records stronger average motion in the given regions.

Final Answer

Method B captures stronger motion.

Justification

Method B has an average motion magnitude of **9 pixels/frame**, while Method A has an average of **5.33 pixels/frame**. Since Method B has the higher average magnitude, it captures stronger motion according to the given data.

QUESTION 5 – PATTERN RECOGNITION ACCURACY COMPARISON

Given Data

System	Correct Predictions	Total Samples
Traditional	85	100
Deep Learning	92	100

(a) Calculate Accuracy

Formula

Traditional System

Deep Learning System

Comparison

System	Accuracy
Traditional	85%

Deep Learning 92%

(b) Evaluation

Difference in accuracy:

The Deep Learning system correctly recognizes more samples than the Traditional system.

Final Answer

Deep Learning performs better.

Justification

The Traditional system achieves **85% accuracy**, whereas the Deep Learning system achieves **92% accuracy**. Since Deep Learning has a **7 percentage-point higher accuracy**, it demonstrates better pattern recognition performance for the given data.

QUESTION 8 – OPTICAL FLOW CONSISTENCY

Given Data

Frame Method A Method B

1	4	6
2	5	8
3	6	9

(a) Calculate Average Motion

Method A

Method B

(b) Compare Consistency

To compare stability using the given three values, we can compare their **range**.

Method A

Method B

Method A has the smaller range and therefore shows less variation across the three frames.

Comparison

Method	Average Motion	Range	Stability
Method A	5.00	2	More stable
Method B	7.67	3	Less stable

Final Answer

Method A is more stable and consistent.

Justification

Method A has an average motion of **5 pixels/frame** and a smaller range of **2 pixels/frame**. Method B has a higher average motion of **7.67 pixels/frame**, but its range is **3 pixels/frame**. Therefore, based on variation across the given frames, **Method A is more consistent and stable**.

QUESTION 9 – DETECTION ACCURACY COMPARISON

Given Data

	Model Correct Predictions	Total Samples
A	180	200
B	170	200

(a) Calculate Accuracy

Formula

Model A

Model B

Comparison

Model	Accuracy
Model A	90%
Model B	85%

(b) Evaluation

Difference:

Model A correctly identifies more samples than Model B.

Final Answer

Model A performs better.

Justification

Model A achieves **90% accuracy**, whereas Model B achieves **85% accuracy**. Since Model A has a **5 percentage-point higher accuracy**, Model A demonstrates better detection performance based on the given data.

QUESTION 11 – MOTION ESTIMATION VARIANCE

Given Data

Frame Method A Method B

1	2	6
2	3	7
3	2	5

(a) Calculate Mean Error

Method A

Method B

(b) Compare Variability and Stability

Using the range of the given errors:

Method A

Method B

Method A has both a lower mean error and lower variation.

Comparison

Method	Mean Error	Range	Stability
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Method A	2.33 pixels	1	More stable
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Method B	6.00 pixels	2	Less stable
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Final Answer

Method A is more accurate and stable.

Justification

Method A has a mean error of only **2.33 pixels**, compared with **6 pixels** for Method B. Method A also has a smaller range of error. Therefore, Method A produces **lower errors and more consistent results**, making it the better method based on the given data.

QUESTION 12 – IOU COMPARISON

Given Data

Object Method A Method B

1	0.60	0.80
2	0.70	0.85
3	0.65	0.90

(a) Calculate Average IoU

Formula

Method A

Method B

Comparison

Method Average IoU

Method A 0.65

Method B 0.85

(b) Evaluation

Difference:

Method B has a substantially higher average IoU.

IoU represents the overlap between the predicted region and the actual region. Therefore, a higher IoU indicates better localization quality.

Final Answer

Method B provides better detection quality.

Justification

Method B achieves an average IoU of **0.85**, compared with **0.65** for Method A. The higher IoU indicates better overlap between the predicted and actual object regions. Therefore, **Method B provides better object localization and detection quality.**

QUESTION 13 – RECALL COMPARISON IN DETECTION

Given Data

Model True Positives (TP) False Negatives (FN)

A 60 40

B 80 20

(a) Calculate Recall

Formula

Model A

Model B

Comparison

Model TP FN Recall

Model A 60 40 60%

Model B 80 20 80%

(b) Evaluation

Difference:

Model B also has fewer false negatives:

- Model A = **40 FN**
- Model B = **20 FN**

Therefore, Model B misses fewer actual positive cases.

Final Answer

Model B has better sensitivity and recall.

Justification

Model B achieves a recall of **80%**, while Model A achieves only **60%**. Model B also has fewer false negatives, meaning it detects a larger proportion of the actual positive cases. Therefore, **Model B provides better sensitivity and detection performance.**